Module-04 **RBT Level-**L1, L2

Introduction to Power View- Introduction to power view, Basic Visualizations and Visual interactions, Filters and Drill Thru Reports, Hierarchies and Drill Down Reports/Visualizations,Grouping, Binning and Sorting, Bookmarks and Selection pane

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Introduction to power view

Power View is an interactive data visualization feature used in Microsoft Power BI. It allows users to create dynamic, visually rich reports and dashboards from their data without needing advanced programming skills.

Key Features of Power View

Interactive Visualizations

- Create charts, tables, maps, and cards
- Click on elements to filter and explore data dynamically

Drag-and-Drop Interface

- Easy to build reports by simply dragging fields onto the canvas

Multiple Chart Types

- Bar charts, column charts, line charts, pie charts, scatter plots, maps, etc.

Data Exploration

- Enables quick analysis through filtering, slicing, and highlighting

Dashboard Creation

- Combine multiple visuals into a single report page for storytelling

Integration with Data Models

- Works with data models created using Power BI or Excel Power Pivot

Components of Power View

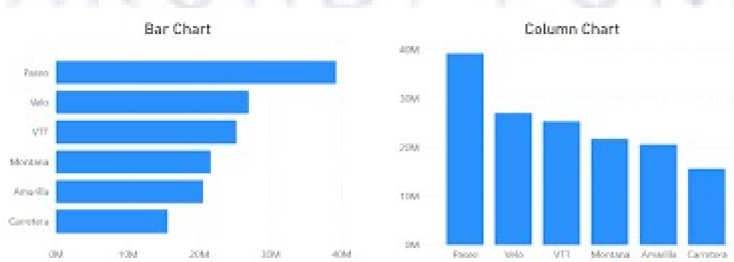
- **Fields Pane** – Select data fields to use in visuals
- **Visualization Canvas** – Area where reports are created
- **Filters Pane** – Apply filters to control data view
- **Design/Layout Options** – Customize appearance and format

Advantages of Power View

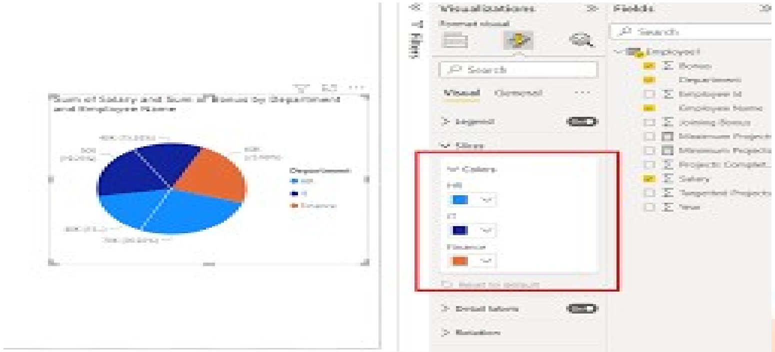
- User-friendly and no coding required
- Real-time interaction with data
- Supports storytelling through visuals
- Helps in quick decision-making

Basic Visualizations and Visual interactions

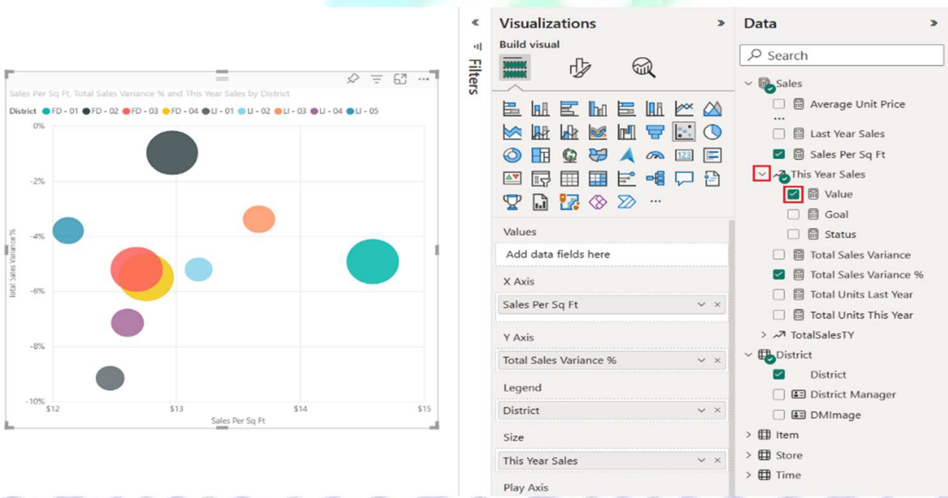
- **Tables:**
    - In table we present data in 2-Dimension data grid format.
    - By default display all data, means flat data structure.
    - Rows not fixed but columns are fixed.
    - When you add new dimension you can add them as “Values” and it will appear as a new column in grid.
- **Matrices:**
    - It support Multidimensional data grid format.
    - Matrix automatically aggregate the data according to data behavior.
    - Rows & Columns both are not fixed.
    - Here we have option to add Values, Rows & columns.
- **Bar & Column Charts:** Bar and Column charts in Power BI are essential for comparing categories, using horizontal bars (Bar) or vertical columns (Column) to represent data magnitudes.
    - **Clustered Column/Bar Chart:** Compares multiple measures side-by-side.
    - **Stacked Column/Bar Chart:** Shows total values while breaking down contributions by segment.
    - **Key Differences:**
        - Bar Chart: Horizontal, ideal for long labels.
        - Column Chart: Vertical, better for chronological time comparison.



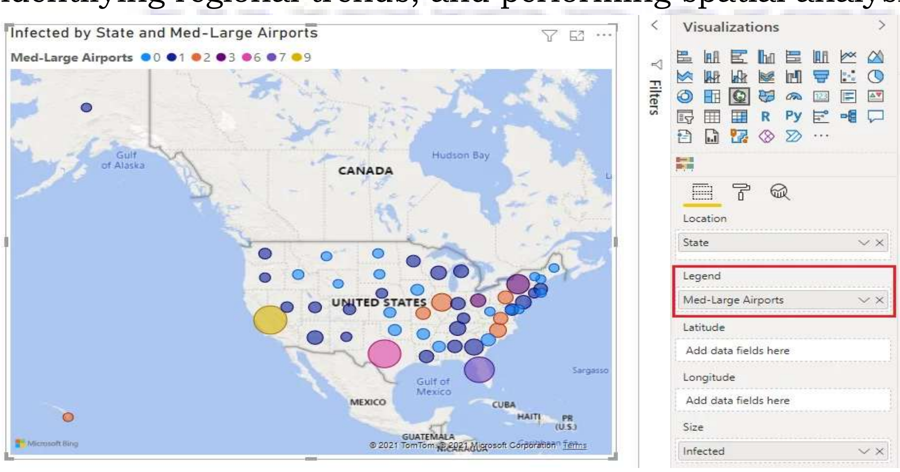
❖ **Pie Charts:** Used for showing parts-to-a-whole. Power View allows for sophisticated, nested pie charts.



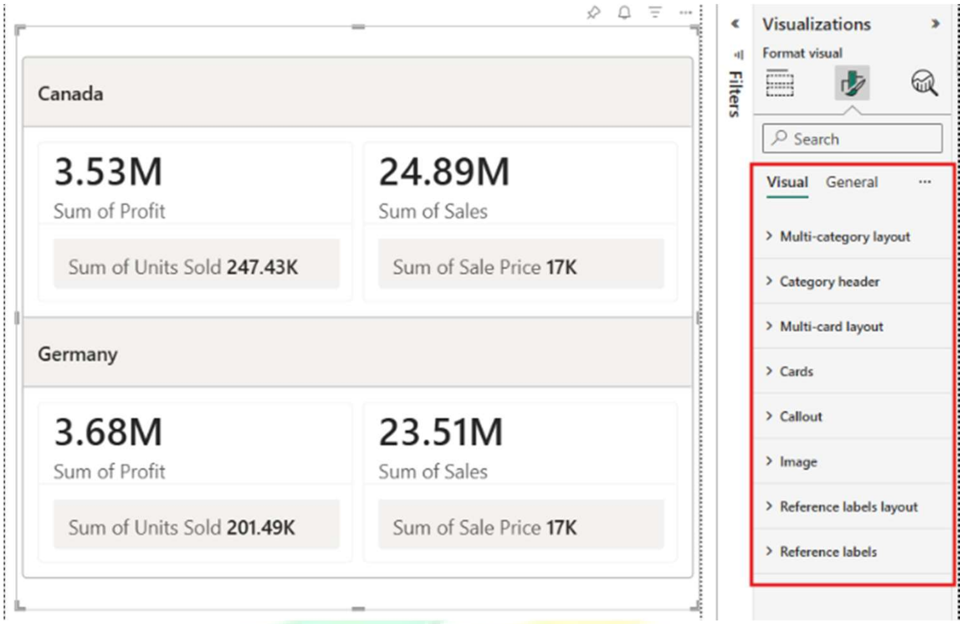
❖ **Scatter & Bubble Charts:** Scatter and bubble charts in Power BI are powerful visuals used to identify relationships, correlations, clusters, and outliers in numerical data.



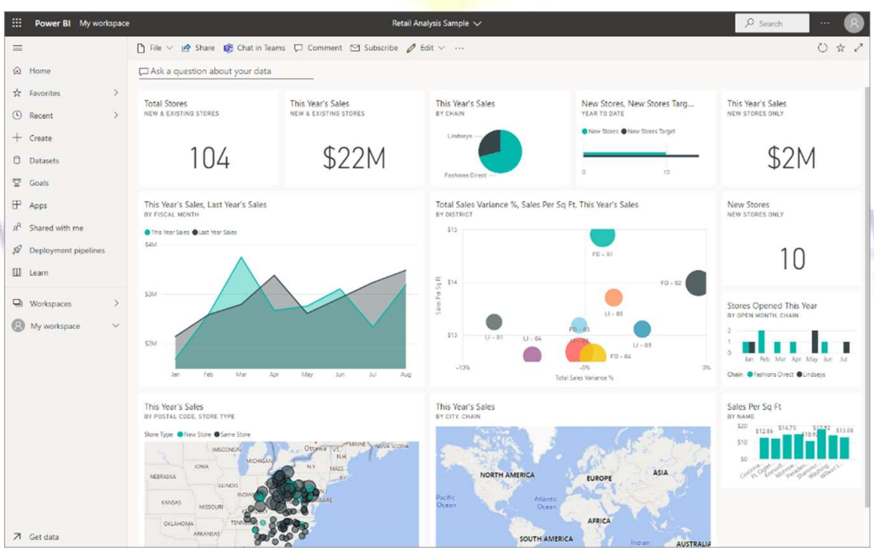
❖ **Maps:** Maps in Power BI are essential for visualizing geographic data, identifying regional trends, and performing spatial analysis.



❖ **Cards:** Cards in Power BI are one of the most frequently used visuals, designed to highlight a single, critical data point such as total sales, count of customers, or average growth in a clear and unmissable format.



❖ **Tiles:** It is a snapshot of a report visual pinned to a dashboard, serving as a high-level container for key metrics. Tiles can display data from reports, dashboards, Excel, or streaming sources.



Power BI - Drilling Down and Up in Hierarchies

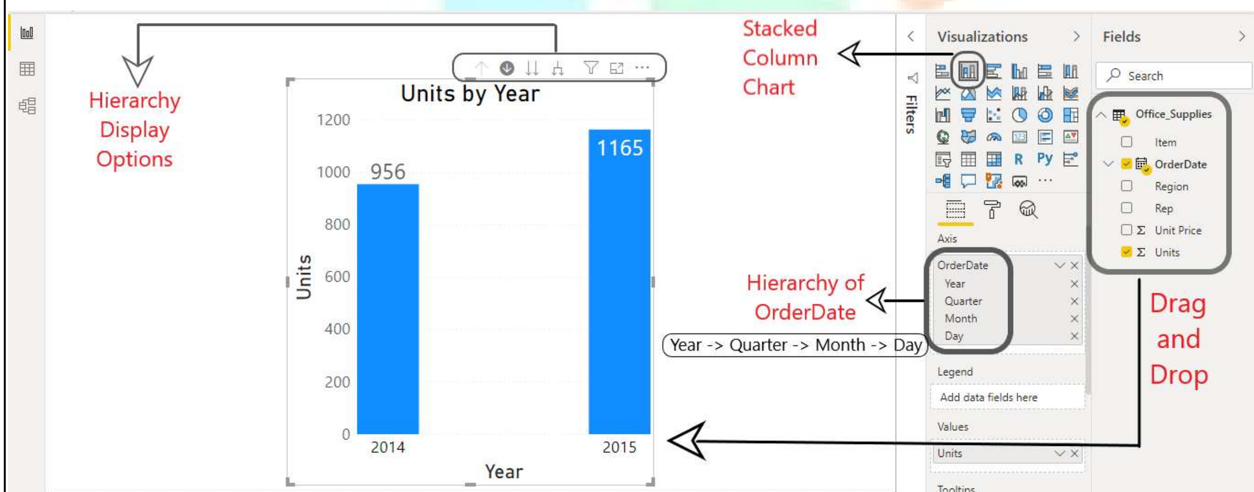
Drilling enables you to navigate through different levels within the dimensional hierarchy of your data source. This allows you to review underlying data for a particular area, and move through the structure of your data source based on your informational needs. Drilling can be performed in two ways:

- Drill up
- Drill down

Dataset Used: The dataset used is of '**Office Stock Supplies**'.

Before getting to understand how Drilling works, we need to create a bar chart.

Creating a Bar Chart:



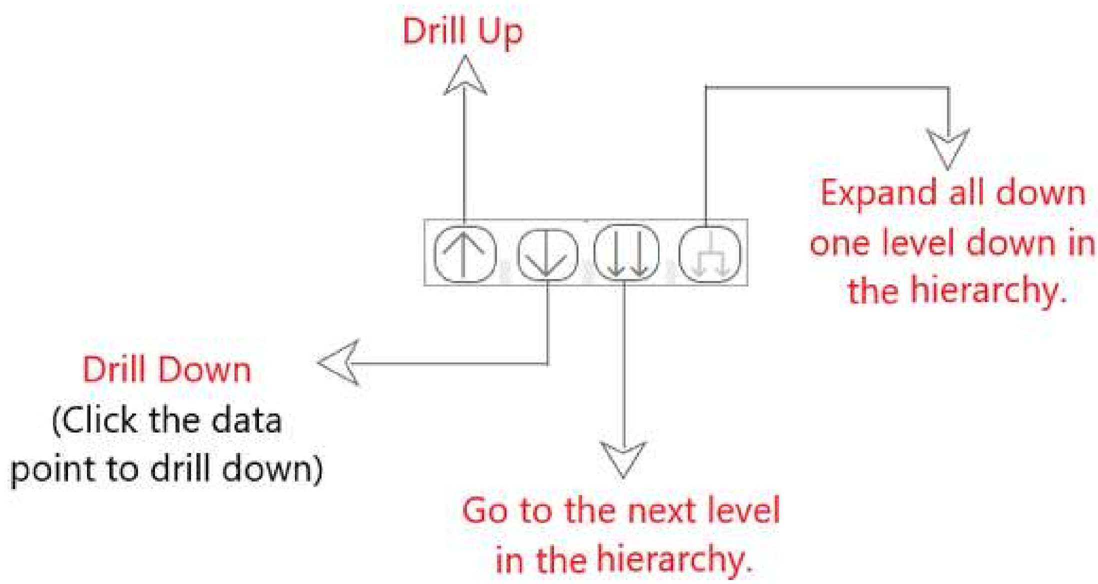
To create a Bar Chart (stacked column chart), do the following steps :

Step 1 - Upload your dataset into your Power-BI Model.

Step 2 - Drag columns [order date, Units] from Fields section.

Step 3 - Click on the Stacked Column Chart in the Visualization Panel.

Hierarchy Display Panel:



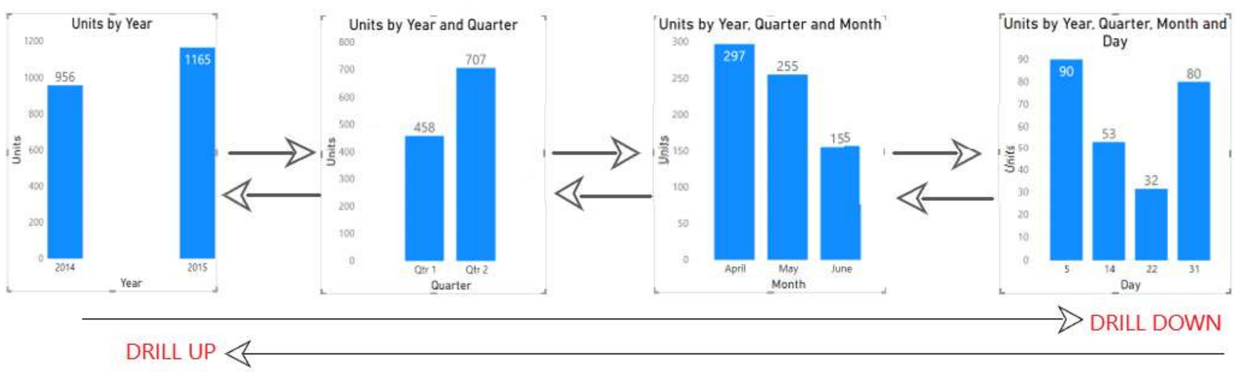
Drilling in Hierarchies: Drilling down allows us to get a granular view of the data in the datasets and we use Drill up to get back the original data.

Step 1 - Click on the drill down icon to turn it ON.

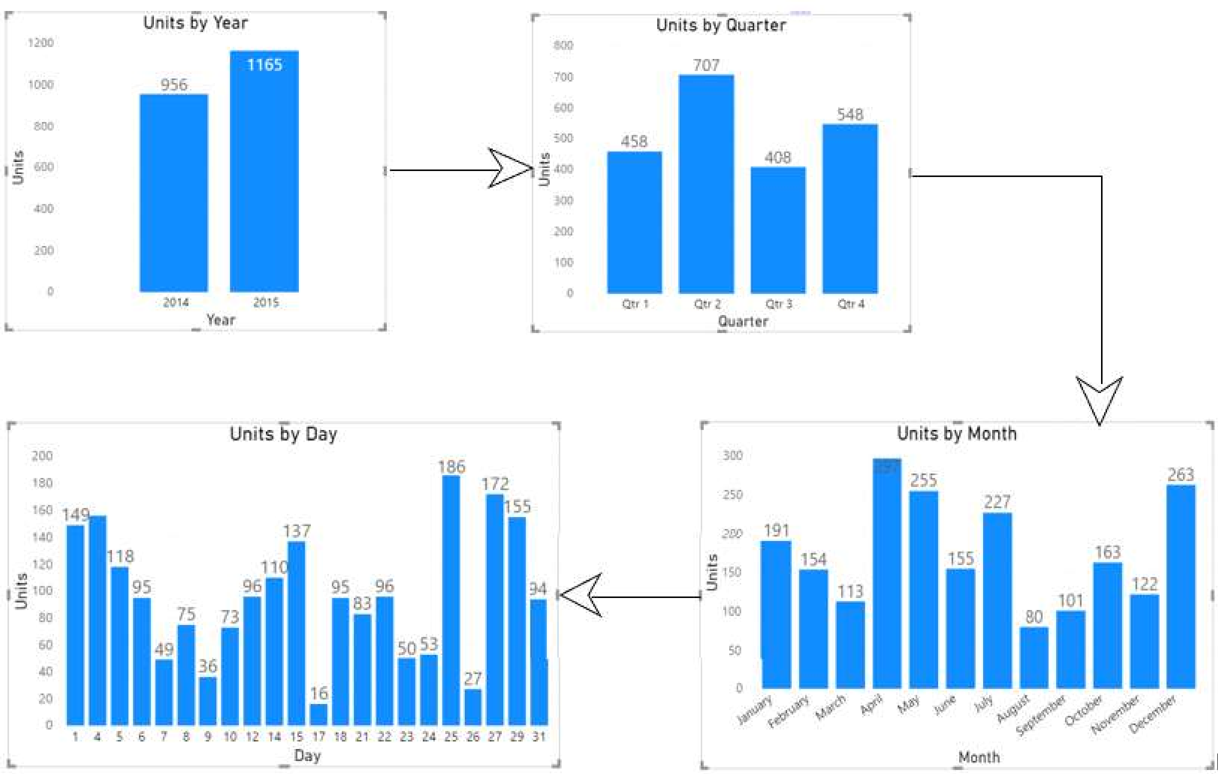
Step 2 - Click on any of the column bars (say column 2) to Drill it down.

(This will show the data of that particular column only)

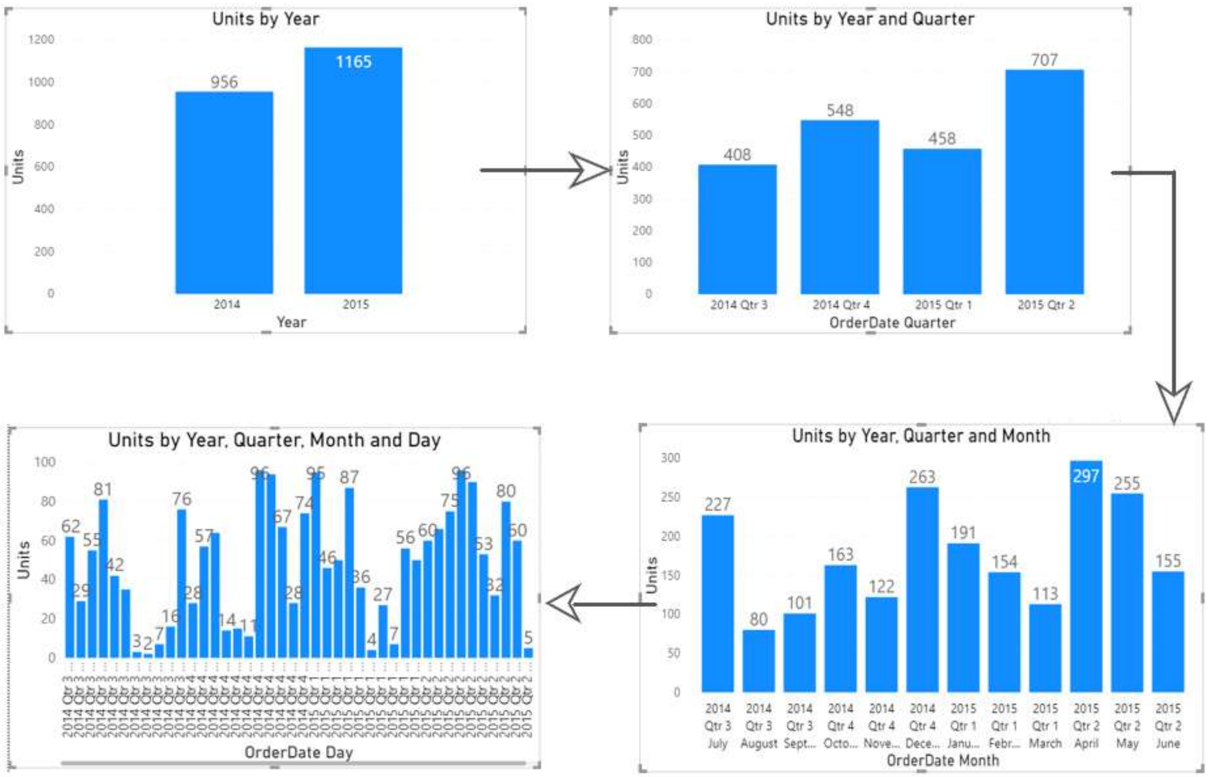
Example : Year (2015) --> Quarter (2) --> Month (May) --> Day (Days in month of May only) [Shown in image below]



Advanced Drilling in Hierarchies - Switch to Next Level In this, drilling is done on the entire data, rather than just a particular section of data. Higher-level hierarchies are not preserved when switching to next level. In simple terms, on switching to the next level, say from year to quarter we cannot see which quarter belongs to which year. As per the hierarchy of OrderDate [Year --> Quarter --> Month --> Day], on clicking this button, Power BI completely disregards which year we want to drill into, and shows all the data available in Quarterly format. See in the image given below, how data is shown after switching to each next level.



Expand down one level: Unlike the previous approach, here on expanding to the next level, the data of the previous level is preserved. (i.e. on switching to the next level, from year to quarter we can see which quarter belongs to which year.) The information of the previous levels is preserved in the shown level.



Hierarchies

A hierarchy is a structured way of organizing related fields in levels from highest to lowest so users can explore data step-by-step.

👉 Example:

- **Date Hierarchy**
  - Year → Quarter → Month → Day

👉 Example:

- **Location Hierarchy**
  - Country → State → City

Why Use Hierarchies?

- Enables **Drill Down / Drill Up** in visuals
- Improves **data exploration and storytelling**
- Makes reports more **interactive and user-friendly**
- Organizes complex datasets into meaningful levels

🔷 How to Create Hierarchy in Power BI

☑ Steps:

- ➢ Go to the Fields Pane (right side of Power BI Desktop)
- ➢ Right-click on a column (e.g., Country)
- ➢ Click "Create Hierarchy"
- ➢ A new hierarchy will be created with that field as the first level
- ➢ Drag and drop other fields (e.g., State, City) into the hierarchy
- ➢ Rename it if needed (e.g., Location Hierarchy)

👉 Final structure:

- Country
  - State
    - City

There are 2 types of Hierarchy in PBI

- ❖ System Defined Hierarchy
- ❖ User defined Hierarchy

1)System Defined Hierarchy:

- ❖ created by PBI
- ❖ Where ever the date column the hierarchy will be create automatically.
- ❖ This Hierarchy can’t be modified.

2)User defined Hierarchy

- ❖ Created by the developer

Create Hierarchy.

Go Power view → Go product table → Select CategoryName → Row Context on it →

← Rename this Hierarchy as ProdHierarchy ← Select create Hierarchy ←

→ Select SubCategoryNameRc on it → Add to Hierarchy → Take the name of ProdHierarchy → Select ProductName Rc on it → Add to Hierarchy → Take the name of ProdHierarchy.

Drill Down:

It will display

ProdHierarchy
CategoryName
SubCategoryName
ProductName

- Go new Page,select stacked column chart place the ProdHierarchy and SalesAmount.
- In previous we create visualization

1)Category Name with SalesAoumt

2)SubCategory Name with SalesAoumt

3)ProductName with SalesAoumt

- Here all Drill Down Options will appear automatically Right side top Corner.
    - Drill Up
    - Click to Turn on Drill Down
    - Go To the Next level in the Hierarchy
    - Expand all down one level in the Hierarchy
- When Click on all these options we see the different category of data.
- when click on Drill Up, it display the SalesAmount by CategoryName
- Click on Next level in the Hierarchy then it will display SalesAmount by SubCategoryName
- Click on Next One then it will display SalesAmount by ProductName
- Create stacked column chart visualization place the OrderDate and SalesAmount.
- When click on lowest level of data arrows(It Drill Down) it will display the data Year,Month,Day wise data.
- when click on Drill Up that will back.

Grouping in PowerBi

Grouping in Power BI is a data modeling and visualization feature that combines multiple discrete values (text or numbers) into a single, higher-level category or bucket.

When working with extensive datasets in Power BI, it's essential to have a strategy to group and categorize information effectively. The process of creating groups in Power BI, making data more manageable and user-friendly.

1. **Access the Data Pane:** Start by opening Power BI project and selecting the Data Pane on the right side of the screen.

2. **Choose Column:** Identify the column want to create groups for.

3. **Right-Click and Select "New Group":** Right-click the chosen column and choose "New Group" from the menu.

4. **Name the Groups:** Give group meaningful name to avoid confusion.

5. **Create Your Groups:** Select the values in the column want to group and click "Group." This action generates new groups based on selection.

6. **Exclude Ungrouped Values:** If there are values that shouldn't be part of any group, uncheck the "Include Other Group" option.

7. **Apply Changes:** Click "OK" to save group.

Binning in PowerBi

Binning is another handy feature that Power BI offers, allowing to group numerical data into ranges, making it easier to analyze.

1. Access the Data Pane: Just like in the grouping process, open Power BI project and navigate to the Data Pane.

2. Select the Column: Locate the numerical column want to create bins for.

3. Right-Click and Choose "New Group": Right-click the column and select "New Group."

4. Create Bins: In this step, we'll define the bin size or number of bins.

5. Apply Changes: Once we've set bin parameters, click "OK."

Bookmarks and Selection pane

A bookmark captures the state of a report page. It includes the changes you made to filters, slicers, and visuals on that page.

There are two types of Power BI bookmarks:

- Personal bookmarks
- Report bookmarks.

Personal Bookmarks are user-specific saved views of a Power BI report that are not shared with others.

Report Bookmarks are predefined views created by the report designer and shared with all users of the report.

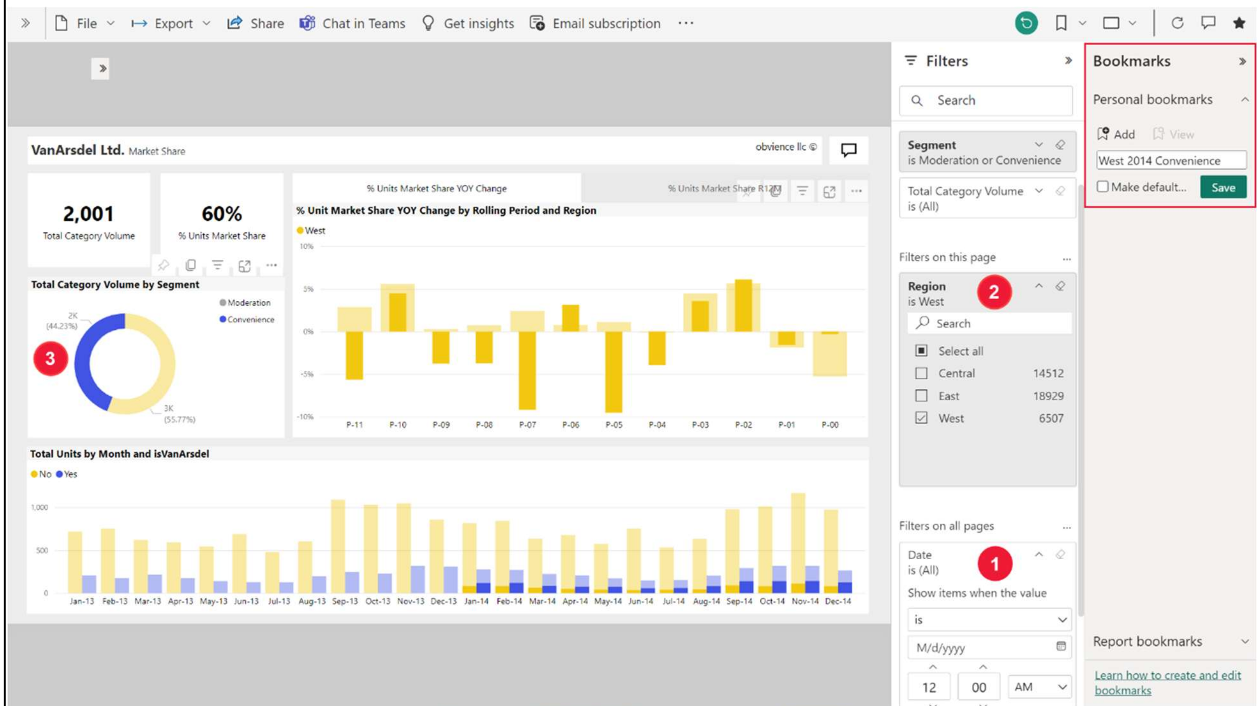
Create personal bookmarks in the Power BI service

If we can view a report, then we can also add personal bookmarks. The maximum number of personal bookmarks per report is 20. When we create a bookmark, the following elements are saved with the bookmark:

- The current page
- Filters
- Slicers, including slicer type (for example, dropdown or list) and slicer state
- Visual selection state (such as cross-highlight filters)
- Sort order
- Drill location

Create personal bookmark.

Configure a report page the way we want it to appear in the bookmark. The following example has filters applied to the default report page:



1. Date is changed to include All dates.
2. Region filter is changed to include only West .
3. A specific data point on the donut chart visual is selected. This selection cross-filters and cross-highlights the other visuals on the report canvas.
4. After your report page and visuals are arranged how you want them, select the bookmarks icon and choose **Show more bookmarks** to open the **Bookmarks** pane.
5. From the **Bookmarks pane**, select **Personal bookmarks > Add.**
6. The personal bookmark gets a generic name, or you can enter a name. If you want this bookmarked view to be your default view, select the **Make default** check box.
7. Select Save. To edit your bookmark, select the ellipses next to the bookmark's name and choose **Update, Make default, Rename, or Delete.**

S.No	Questions
01	**Define** Power view? **List** and **explain** key features and task Transformed in Power view.
02	**Explain** Components of Power View? mentation Advantages of Power view.
03	**List** and **explain** Basic Visualizations.
04	**Explain** Drilling Down and Up in Hierarchies
05	**Define** Hierarchies? Explain Why Use Hierarchies
06	**List** and **explain** how to Create Hierarchy in Power BI
07	**Explain** Drill Down report.
08	**Write** short note on 1) Grouping in PowerBi 2) Binning in PowerBi 3) Bookmarks